Introduction to biology

01 / What is life?

Characteristics of living systems

Living systems have cellular organization, use energy, regulate internal conditions, respond to stimuli, grow and develop, and reproduce. Populations evolve across generations. Individual organisms can be alive even when they cannot reproduce; these characteristics describe living systems rather than a rigid checklist for each individual.

Homeostasis

Homeostasis is the regulation of internal conditions within a functional range despite changes inside or outside the organism. Negative feedback counteracts a deviation from a regulated condition. Sweating supports temperature regulation because evaporation transfers heat away from the skin, helping cool an overheated body.

Prokaryotic and eukaryotic cells

Prokaryotic cells lack a membrane-bound nucleus; their DNA occupies a nucleoid region. Eukaryotic cells have a membrane-bound nucleus and other membrane-bound organelles. Both kinds of cells have a plasma membrane, cytoplasm, genetic material and ribosomes.

Levels of organization

An organism is an individual living entity. In multicellular organisms, specialized cells can form tissues, tissues form organs, and organs cooperate in organ systems. A unicellular organism carries out life's functions within one cell and does not have this tissue-organ hierarchy.

Original mock course material for a portfolio demo. Not an institutional handout.

Introduction to biology

02 / Asking biological questions

Hypotheses and predictions

A hypothesis is a testable proposed explanation for an observation. A prediction states an expected observation if the hypothesis and relevant assumptions are correct. A hypothesis about light and plant growth can predict that otherwise comparable plants receiving more light will grow taller.

Independent and dependent variables

The independent variable is deliberately changed by the experimenter, such as the amount of light received by a plant. The dependent variable is the measured response, such as plant height or growth. Keep water, soil and plant type comparable to reduce alternative explanations for the result.

Control groups

A control group provides a baseline for comparison with the treatment group. Holding other conditions comparable helps isolate the effect of the independent variable. Random assignment and suitable controls reduce systematic differences that could otherwise explain the observed outcome.

Repeating experiments

Repeating an experiment helps estimate random variation and reveals whether a result is reproducible. Independent repetitions reduce the influence of a single unusual observation and can reveal anomalies. Repetition alone does not remove a systematic bias; improving controls or methods may also be necessary.

Observation, inference and evidence

An observation records information gathered through senses or instruments. An inference interprets observations using reasoning. Evidence can support or challenge a hypothesis, but a single favorable result does not establish universal certainty. Interpret results together with the study's controls and limitations.

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